

1

NUMBER AND ALGEBRA

INTEGERS

Antarctica is the coldest continent on Earth as it is at the South Pole. During its winter, from March to August, the Sun does not rise, the sky stays dark for 6 months and inland temperatures are from −66°C to −57°C. During its summer, from September to February, the Sun does not set, the sky stays bright for 6 months, inland temperatures are from −45°C to −25°C and this is the only safe time to visit. Antarctica is permanently covered with snow and ice, so it reflects the Sun's rays rather than absorbing them, and its dry air and high altitudes contribute to its sub-zero temperatures.



Chapter outline

	Working mathematically				
1.01 Numbers above and below zero	U	F	PS	R	C
1.02 Integers on a number line	U	F		R	
1.03 Ordering integers	U	F		R	C
1.04 Adding integers	U	F		R	C
1.05 Subtracting integers	U	F		R	C
1.06 Multiplying integers	U	F		R	C
1.07 Dividing integers	U	F		R	C
1.08 Order of operations	U	F		R	C
1.09 Integers and the calculator	U	F		R	C
1.10 Integer problems	U	F	PS	R	

Wordbank

ascending Moving upward, increasing from smallest to largest

descending Moving downward, decreasing from largest to smallest

evaluate To find the alue of an expression

integer A positive or negative whole number or zero

minus To subtract or 'take-away' a number

negative number A number that is less than zero; the opposite of a positive number

number line A line that shows the position and order of numbers

Quiz
Wordbank 1

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (5):

1.04 Adding integers
1.05 Subtracting integers
1.07 Multiplying and dividing integers
1.08 Order of operations • BODMAS

***Twig* video (1):**

1.01 India and negative numbers

***PhET* interactives (3):**

1.02 Number line: Integers
1.04 Number line: Operations
1.05 Number line: Operations • Number line: Distance

Quizzes (6):

- Wordbank 1
- SkillCheck 1
- Mental skills 1A
- Mental skills 1B
- Language of maths 1
- Test yourself 1

Skillsheets (5):

1.02 Spreadsheets
1.04 Integers using coloured squares
1.05 Integers • Integers using coloured squares
1.06 7 Integers
1.08 Order of operations

Worksheets (6):

1.02 A page of number lines
1.05 Adding and subtracting integers
1.06 Integers writing activity
1.07 Integers review • What is the question?
Mind map Integers

Puzzles (7):

1.02 Integer snap
1.04 Integers game
1.06 Multiplying integers game
1.08 Order of operations puzzle • Directed numbers
1.10 Integers group clues
Language of maths Integers find-a-word

Presentations (2):

1.04 Adding directed numbers
1.05 Subtracting directed numbers

Spreadsheet (1):

1.06 Integer quiz: Multiplication

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- understand and apply positive and negative whole numbers and zero, and represent them on a number line
- order, add, subtract, multiply and divide integers, with and without technology
- apply 'order of operations' rules with integers
- solve problems involving operations with integers

SkillCheck ANSWERS ON P. 561

1 Simplify each **expression**.

a $7 + 9$	**b** $12 - 4$	**c** 7×4	**d** $20 \div 5$
e $13 + 8$	**f** $30 - 5$	**g** 2×11	**h** $15 \div 3$
i $6 + 7$	**j** $14 - 6$	**k** 3×5	**l** $24 \div 4$

2 Simplify each expression.

a $12 + 37$	**b** $652 - 48$	**c** 14×5	**d** $66 \div 2$
e 79×3	**f** $384 \div 3$	**g** $3 + 5 \times 4$	**h** $18 \div 2 - 7$
i $30 - 12 \div 6$	**j** $20 \times (3 + 17)$	**k** $12 - 3 \times 2 + 6$	**l** $43 - [5 \times 2 + 1]$
m 3^2	**n** 5^2	**o** 8^2	**p** 1^2

3 State whether each statement is true (T) or false (F).

a $15 > 12$	**b** $8 < 11$	**c** $6 < 3$
d $4 + 3 < 10$	**e** $12 - 3 > 11$	**f** $7 < 3 \times 2$
g $15 \div 3 > 12 \div 4$	**h** $8 + 3 + 2 > 6 \times 1$	**i** $18 \div 3 < 2 + 2 + 2$

4 Write each set of numbers in ascending order (smallest to largest).

a 3, 8, 2, 1, 4	**b** 2, 18, 5, 7, 3, 16	**c** 38, 51, 49, 27, 66, 54

5 Write each set of numbers in descending order (largest to smallest).

a 6, 11, 18, 17, 5, 3	**b** 8, 7, 14, 23, 31, 5, 2	**c** 89, 91, 37, 69, 41, 3, 46

Quiz
SkillCheck 1

Skillsheet
Ordering numbers

Numbers above and below zero

1.01

Video
India and negative numbers

'The temperature in Canberra was 3°C yesterday, but overnight it dropped 5 degrees to –2°C'.

'When Lisa withdrew $400 from her bank account, her bank balance of $340 went down to –$60'.

Both of the above situations involve numbers above and below zero.

Numbers greater than zero, such as 3, 400, $8\frac{1}{2}$ and 14.37, are called **positive numbers**.

Numbers less than zero, such as –2, –60, $-10\frac{3}{4}$ and –21.6, are called **negative numbers**.

The number –2 is the number that is 2 below 0. It is called 'negative 2', not 'minus 2'.

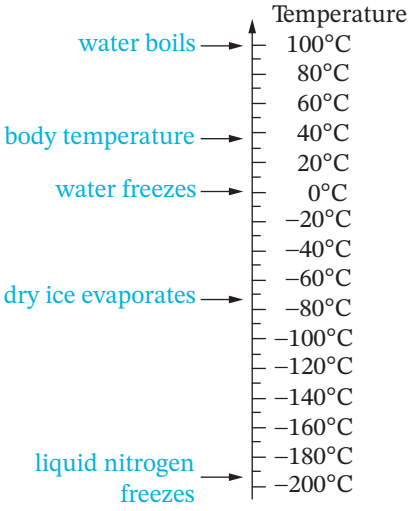
The number 3 is the number that is 3 above 0. It can also be called 'positive 3' and be written as '+3'.

The negative numbers are the opposites of the positive numbers. The opposite of 5 is –5.
The opposite of –9 is 9.

EXERCISE 1.01 ANSWERS ON P. 561

Numbers above and below zero U F PS R C

1 This scale shows temperatures in degrees Celsius (°C) above and below 0°C. R C



a At what temperature does water boil?

b What is measured at 37°C?

c Estimate the temperature at which dry ice evaporates.

d What is measured at 0°C?

e Last week, it was 48°C in Algeria, Africa, and −69°C at the South Pole. What was the difference between the 2 temperatures?

2 For each number, write its opposite number.

a 6 **b** 1 **c** −10

d 11 **e** 0 **f** −2

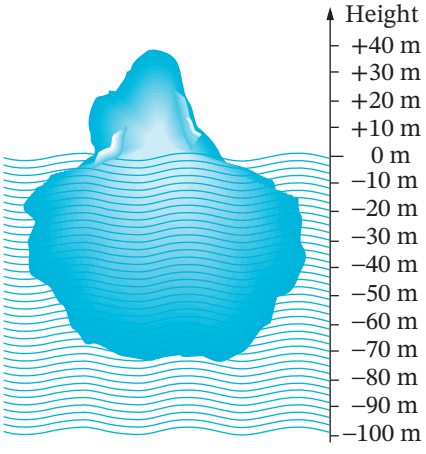
3 Write a word that is the opposite of: C

a up **b** more **c** down

d left **e** before **f** south

g withdraw **h** decrease **i** west

4 Icebergs can be dangerous to ships because about $\frac{5}{6}$ of their mass lies under water. The following diagram shows the position of an iceberg above and below sea level. R C



a What is the height of the iceberg above sea level?

b What is the depth of the iceberg below sea level?

c What is the total height of the iceberg?

d Why are '+' and '−' signs useful in these measurements?

5 Dilshan's bank account has money coming in and going out over one day. Withdrawals have a negative sign. PS

Balance at the start:	$720
	−$50
	$60
	−$40
	$100
	$80
	−$260
Balance at the end of the day:	$

Calculate the balance at the end of the day.

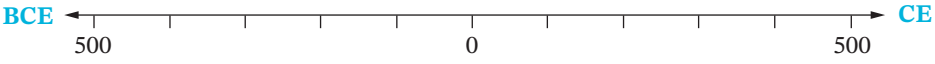
6 Write the opposite of: R C

a withdrawing $25 from the bank
b winning $12
c walking 2 steps backwards
d lowering the temperature by 3°
e driving 5 km west
f increasing the price by $8
g going up 3 floors
h 5 seconds before take-off

7 What number's opposite is itself? R

8 This timeline starts at 500 BCE and ends at 500 CE. R C

CE (or AD) stands for Common Era while BCE (or BC) stands for Before the Common Era.



Copy the timeline and place each event below in the correct place. PS

A 100 CE — The first Chinese dictionary was compiled.
B 400 BCE — The first temple dedicated to the Greek god Zeus was built.
C 490 BCE — Greek runner Phidippides inspired the first marathon with a 26-mile run.
D 55 BCE — Roman forces under Julius Caesar invaded Britain.
E 190 CE — The abacus was invented.
F 370 CE — Hypatia, a female mathematician, was born in Alexandria, Egypt.
G 410 CE — Roman forces left Britain.
H 330 BCE — Greek mathematician Euclid showed that an infinite number of prime numbers exist.
I 260 CE — The 2-year war between Rome and Persia ended.
J 240 BCE — Greek mathematician Eratosthenes estimated the circumference of the Earth using 2 sticks.

9 In the Jewish calendar, the year of Creation, 3760 BCE, is Year Zero, while in the Indian calendar, 78 CE is Year Zero. R

a In what year were you born, according to the:

i Jewish calendar? **ii** Indian calendar?

b What is this year, according to the:

i Jewish calendar? **ii** Indian calendar?

Did you know?

Tutankhamen, King of Egypt

In 1922, English archaeologist Howard Carter discovered the tomb of a pharaoh (king) in the Valley of the Kings in Egypt. The pharaoh's name, Tutankhamen, meant 'living image of Amun', an ancient Egyptian god. Tutankhamen became pharaoh at the age of 9 after the death of his parents, and he died aged 18 in 1323 BCE. The cause of his death remains a mystery – many possibilities have been suggested, including murder, but none can be proved. His tomb contained over 3000 treasures, many of them covered in gold, including his coffin. Carter's discovery was significant because Tutankhamen's tomb was so well-preserved. It still exists in the Valley of the Kings today.



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In what year did Tutankhamen become king?

How many years after his death was King Tutankhamen's tomb discovered?

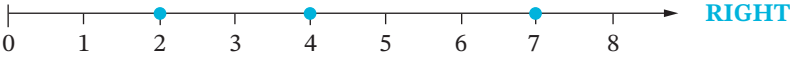
1.02 Integers on a number line

Worksheet
A page of number lines

Puzzle
Integer snap

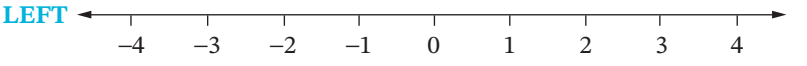
Interactive
Number line: Integers

A **number line** shows the position and order of numbers. The number line below shows the positions of 2, 4 and 7 marked in blue.



As we move further to the right on the number line, the numbers become bigger. The arrow on the right end of the number line shows that numbers can increase infinitely ('forever').

We can extend our number line to the left to include the negative numbers below zero.



As we move further to the left on the number line, the numbers become smaller. The arrow on the left end of the number line shows that numbers can decrease infinitely.

Positive and negative numbers have **size** and **direction**. For example, 5 (or +5) means 5 units in the positive direction, while –2 means 2 units in the negative direction.

Positive and negative **whole** numbers are called **integers**, for example, –3, 4 and –20. Zero is also an integer.

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Integers

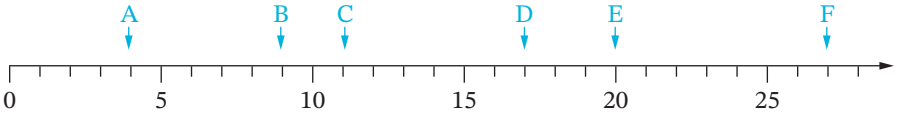
Integers are the positive and negative whole numbers and zero.

Integers are **whole** numbers, not fractions or decimals, so $8\frac{1}{2}$, 14.37, $-10\frac{3}{4}$ and −21.6 are not integers. The general name for **all** positive or negative numbers is the **directed numbers**.

EXERCISE 1.02 ANSWERS ON P. 561

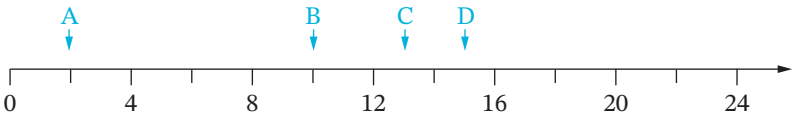
Integers on a number line U F R

1 Write the positive integer indicated by each arrow on this number line.

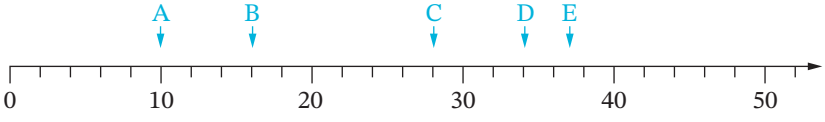


2 Estimate the value of the positive integer indicated by each arrow on the number lines below.

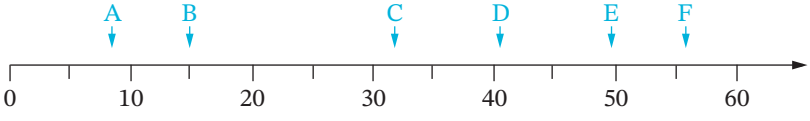
a



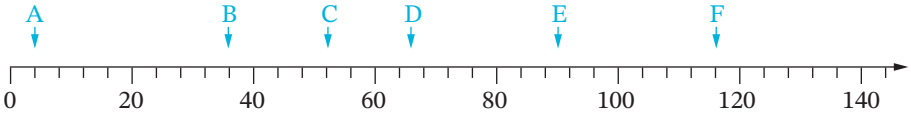
b



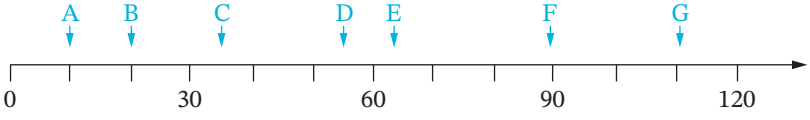
c



d



e

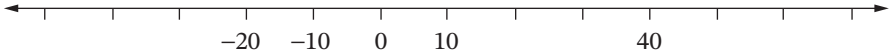


3 Copy and complete each number line using a ruler to mark the positions evenly.

a



b

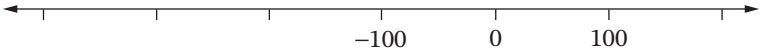


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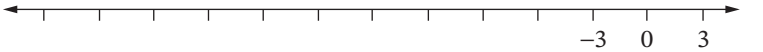
c



d



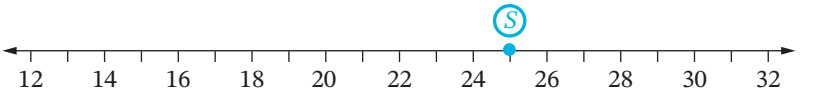
e



4 **a** Draw a number line and show all the integers between −5 and 3.

b Draw a number line marked from −6 to 6 and show all the integers less than 1.

5 On this number line, where will S be if it moves: R



a 4 places to the right?

b 10 places to the right?

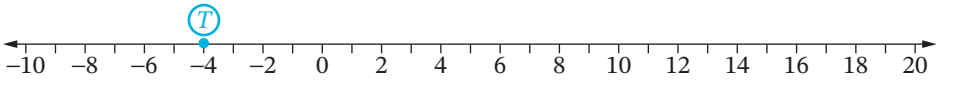
c 7 places to the right?

d 5 places to the left?

e 13 places to the left?

f 20 places to the left?

6 On this number line, where will T be if it moves: R



a 5 places to the right?

b 8 places to the right?

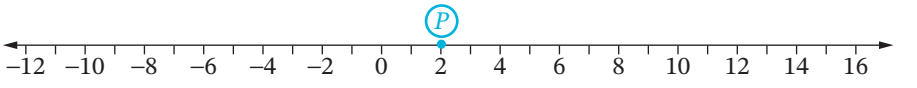
c 10 places to the right?

d 2 places to the left?

e 8 places to the left?

f 11 places to the left?

7 On this number line, where will P be if it moves 4 places to the left and then 9 places to the right? Select the correct answer **A**, **B**, **C** or **D**. R



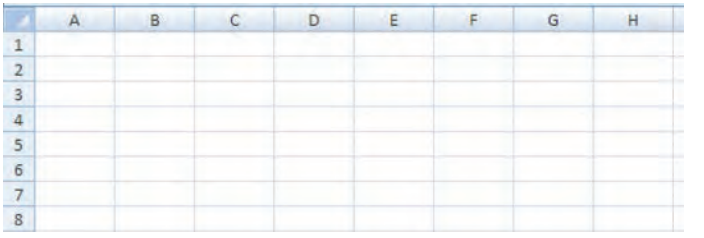
A −11 **B** −3 **C** 7 **D** 15

Skillsheet
Spreadsheets

Technology

Introducing spreadsheets

A spreadsheet is a **calculator** in the form of a grid made up of many cells where we can enter labels and numbers and display answers to calculations. The **columns** of a spreadsheet are ordered A, B, C, D, ..., while the **rows** are ordered 1, 2, 3, 4, ...

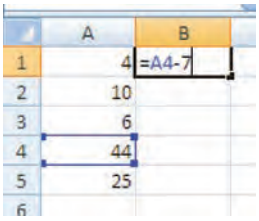


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To make a calculation, we need to write a **formula** (rule) in a cell. A spreadsheet formula uses special symbols and always starts with an equal sign '='. Here are some examples of formulas:

Formula	Meaning
=A1+A2+A3 or **=sum(A1:A3)**	**add** the values in cells A1, A2 and A3
=A5−A4	**subtract** the value in cell A4 from the value in cell A5
=A1*10	**multiply** the value in cell A1 by 10 (* is used instead of ×)
=A1/A2	**divide** the value in cell A1 by the value in cell A2 (/ is used instead of ÷)
=average(A1:A5)	find the **average** of all values from cells A1 to A5



1 Open a new spreadsheet and enter the 5 numbers shown in column A at cells A1, A2, A3, A4 and A5.

2 In cell B1, enter the formula '=A4-7' to calculate the value in A4, minus 7. You should get the answer 37, because 44 − 7 = 37.

3 For the rest of the cells in column B, enter the appropriate formula for each expression shown below and check that the spreadsheet calculates the correct answer. Remember to start each formula with an '=' sign.

a at B2, A2 + A1

b at B3, 10 × A5 − 6

c at B4, A4 − A1 × 2

d at B5, A1 × A2 + A3

e at B6, A1 × (A2 + A3)

f at B7, 3 × A2 ÷ 2

g at B8, $\frac{A5 - A3}{3}$

h at B9, sum of A1, A2, A3, A4, A5

i at B10, average of A1, A2, A3, A4, A5

j at B11, $\frac{A4}{A1} - \frac{A5}{A3}$

4 Now enter **new** values in cells A1 to A5. Using the formulas entered from question **3**, check the new answers calculated in column B.

Ordering integers

1.03

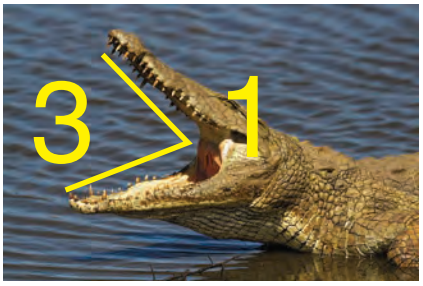
Greater than and less than

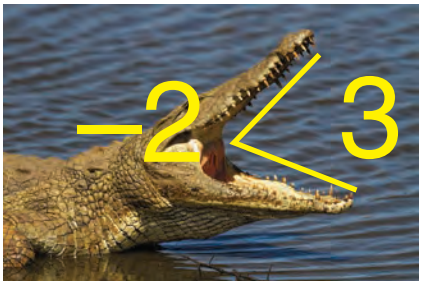
> means 'is greater than'

< means 'is less than'

Think of the > and < signs as being like the mouths of crocodiles. They always open towards the **bigger** number.

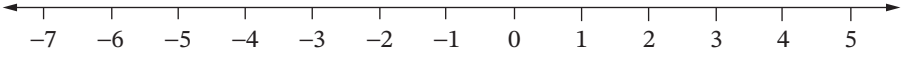
'3 is greater than 1' is written as '3 > 1'. '−2 is less than 3' is written as '−2 < 3'.





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Some more examples are shown in the table below. Remember that on a number line, the numbers on the right are larger than the numbers on the left.



Expression	Meaning	On a number line ...
3 > −4	3 is **greater than** −4	3 is to the **right** of −4
−5 < −3	−5 is **less than** −3	−5 is to the **left** of −3
−6 > −7	−6 is **greater than** −7	−6 is to the **right** of −7
−5 < 2	−5 is **less than** 2	−5 is to the **left** of 2

Example 1

Write these integers in ascending order: −5, 7, 0, 6, 3, −2, 4, −1.

Solution

Ascending order means from smallest to largest.

−5, −2, −1, 0, 3, 4, 6, 7.

Ascending and descending

- **Ascending** means moving UP, from smallest to largest: think of 'A' as a ladder and going up a ladder
- **Descending** means moving DOWN, from largest to smallest: think of 'D' for 'down'

EXERCISE 1.03 ANSWERS ON P. 561

Ordering integers UFRC

1 Copy and complete each statement using a > or < symbol. R C

a 4 ____ 1	**b** 0 ____ 9	**c** 7 ____ −7	**d** 0 ____ −3
e 5 ____ −11	**f** −1 ____ −6	**g** −12 ____ 8	**h** −10 ____ −2
i −6 ____ 5	**j** −6 ____ −2	**k** 17 ____ 23	**l** 24 ____ −4
m −15 ____ −35	**n** −27 ____ 29	**o** −8 ____ 47	**p** −12 ____ 0

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EXAMPLE 1

1.03

2 Here is a list of integers: 1, –2, 3, –4, 5, –6, 7, –8. R C

a Which is the biggest integer?

b Which is the smallest integer?

c Rewrite the integers in order from smallest to biggest.

3 Rewrite each set of integers in ascending order. R C

a –3, 2, –1, 3

b 5, –5, 2, –8, –3

c –4, –6, –3, –10, 0

d 6, –3, 4, –2, –5

e –48, 36, –24, 8, 0, –11

f 15, 12, –10, –26, 3, –2

4 Rewrite each set of integers in descending order. R C

a 4, 3, –1, 5, –2

b –4, 8, –7, –2, 0

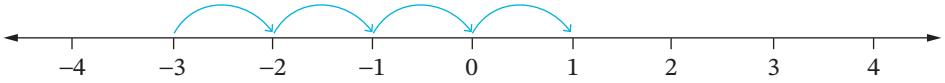
c 1, –1, 4, –5, –11, –3

d 8, –4, –18, 3, –2

e –6, –15, –48, –3, 1

f 33, 1, –100, –58, –36

5 The distance between –3 and 1 on a number line is 4.



Find the distance between each pair of integers.

a 2 and 4

b –2 and –4

c 0 and 4

d 0 and –3

e –3 and 3

f –10 and 10

g –6 and 1

h –1 and 6

6 Which of the following statements is true? Select the correct answer **A**, **B**, **C** or **D**. R C

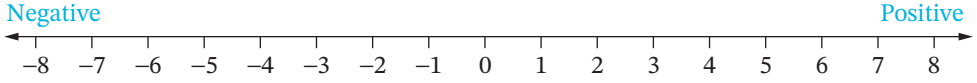
A $-6 > 2$

B $4 < -5$

C $0 < -3$

D $12 > -2$

7 In this question, POS means positive and NEG means negative. R C



a Start at –3. Move 4 steps in the NEG direction. Move 9 steps in the POS direction. Move 6 steps in the NEG direction. Where are you now?

b Start at 8. Move 12 steps in the NEG direction. Move 5 steps in the POS direction. Move 7 steps in the NEG direction. Where are you now?

c Start at –5. Move 6 steps in the POS direction. Move 4 steps in the NEG direction. Move 7 steps in the POS direction. Where are you now?

d Start at 3. Move 5 steps in the POS direction. Move 8 steps in the NEG direction. Move 2 steps in the NEG direction. Where are you now?

e Start at –1. Move 10 steps in the NEG direction. Move 4 steps in the POS direction. Move 5 steps in the NEG direction. Where are you now?

1.04 Adding integers

Integers can be added using a number line.

Video
Adding integers

Presentation
Adding directed numbers

Puzzle
Integers game

Skillsheet
Integers using coloured squares

Interactive
Number line: operations

Example 2

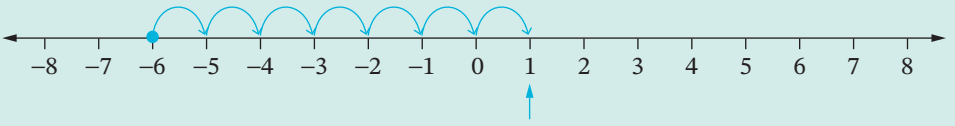
Evaluate each sum.

a $-6 + 7$ **b** $8 + (-6)$ **c** $2 + (-5)$

Evaluate means 'Find the value of'. Here it means 'to work out the answer'. A **sum** is the answer to an addition.

Solution

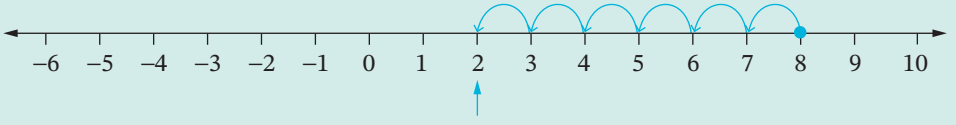
a $-6 + 7$



- −6 is where you start
- + direction is right
- 7 is how far you move

$-6 + 7 = 1$

b $8 + (-6)$



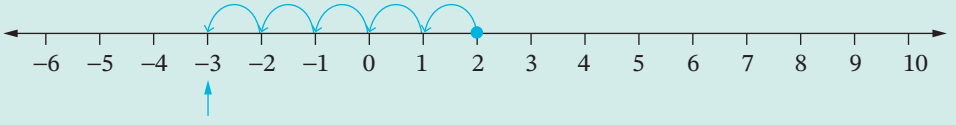
- 8 is where you start
- + direction is right
- −6 is how far you move in the *opposite* direction (that is, left)

$8 + (-6) = 2$

$8 + (-6)$ is the same as $8 - 6$.

Adding a negative integer is like subtracting its opposite.

c $2 + (-5)$



- 2 is where you start
- + direction is right
- −5 is how far you move in the opposite direction (that is, left)

$2 + (-5) = -3$

$2 + (-5)$ is the same as $2 - 5$.

9780170477963

Adding integers

When adding integers on a number line:

- adding a **positive** integer means moving to the right
- adding a **negative** integer is the same as **subtracting its opposite**, so it means moving to the left: $2 + (-5) = 2 - 5$

EXERCISE 1.04 ANSWERS ON P. 561

Adding integers U F R C

EXAMPLE 2

1 Evaluate each sum.

a	$-6 + 10$	**b**	$8 + (-3)$	**c**	$4 + (-5)$	**d**	$3 + (-1)$
e	$-2 + 2$	**f**	$-1 + 5$	**g**	$-7 + 4$	**h**	$-2 + (-2)$
i	$-8 + (-12)$	**j**	$8 + (-12)$	**k**	$-11 + 3$	**l**	$10 + (-10)$
m	$7 + (-9)$	**n**	$-13 + 11$	**o**	$-16 + 5$	**p**	$-7 + 13$

2 What is the correct answer for $-2 + 7$? Select the correct answer **A**, **B**, **C** or **D**.

A 9 **B** -9 **C** 5 **D** -5

3 **a** Is the answer to $-5 + 9$ the same as the answer to $9 + (-5)$?

b Can integers be added in any order? Give an example to support your answer. R C

4 Evaluate each expression.

a	$-6 + (-6)$	**b**	$-11 + 11$	**c**	$-5 + 25$	**d**	$-14 + (-9)$
e	$23 + (-8)$	**f**	$17 + (-17)$	**g**	$0 + (-3)$	**h**	$-27 + 36$
i	$-4 + (-15)$	**j**	$9 + (-19)$	**k**	$-32 + 0$	**l**	$-13 + 4$
m	$3 + (-4) + 1$	**n**	$-1 + 9 + (-3)$				

5 What happens when you add an integer and its opposite? R C

6 For each sum, state whether it is always positive (+), always negative (–), or could be either positive or negative (E). R C

a The sum of 2 positive integers

b The sum of 2 negative integers

c The sum of a positive integer and a negative integer

7 Copy and complete the blank for each equation. R

a	$7 + ___ = 3$	**b**	$-3 + ___ = 4$	**c**	$-3 + ___ = -6$
d	$___ + (-8) = -1$	**e**	$___ + 5 = 2$	**f**	$___ + (-5) = 2$

Foundation Standard Complex

8 Copy and complete the blanks for each equation, where at least one of the numbers must be negative. R

a ___ + ___ = 7 **b** ___ + ___ = −6 **c** ___ + ___ = 0

d ___ + ___ = 1 **e** ___ + ___ = −4 **f** ___ + ___ = −9

9 Copy and complete the blanks for each equation (0s not allowed!). R

a ___ + ___ + ___ = 2 **b** ___ + ___ + ___ = −1

c ___ + ___ + ___ = −3 **d** ___ + ___ + ___ = 0

Quiz
Mental skills 1A

Mental skills 1A: Maths without calculators ANSWERS ON P. 562

Adding 8 or 9

A quick way to mentally add 9 or 8 to a number is to add 10 and count back 1 or 2 respectively.

1 Study each example.

a $17 + 9 = 17 + 10 - 1$
$= 27 - 1$
$= 26$
Count: '17, 27, 26'

b $44 + 8 = 44 + 10 - 2$
$= 54 - 2$
$= 52$
Count: '44, 54, 52'

c $128 + 19 = 128 + 20 - 1$
$= 148 - 1$
$= 147$
Count: '128, 148, 147'

d $256 + 38 = 256 + 40 - 2$
$= 296 - 2$
$= 294$
Count: '256, 296, 294'

2 Now evaluate each sum by counting.

a	146 + 9	**b**	212 + 9	**c**	308 + 9	**d**	1755 + 9
e	29 + 19	**f**	687 + 39	**g**	254 + 29	**h**	933 + 19
i	714 + 8	**j**	623 + 8	**k**	207 + 8	**l**	155 + 8
m	386 + 8	**n**	418 + 48	**o**	909 + 28	**p**	277 + 18

1.05 Subtracting integers

Video
Subtracting integers

Integers can also be subtracted using a number line.

Presentation
Subtracting directed numbers

Example 3

Evaluate each difference:

A **difference** is the answer to a subtraction.

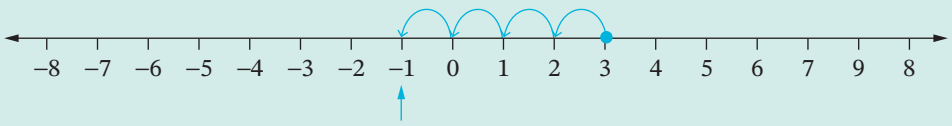
a 3 − 4 **b** −3 − 4 **c** 4 − (−3) **d** −3 − (−1)

Foundation Standard Complex

9780170477963

Solution

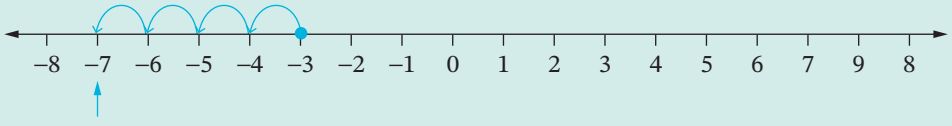
a 3 – 4



- 3 is where you start
- – direction is left
- 4 is how far you move

3 – 4 = –1

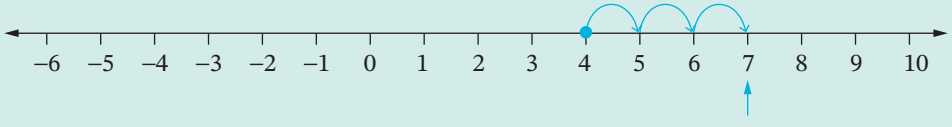
b –3 – 4



- –3 is where you start
- – direction is left
- 4 is how far you move

–3 – 4 = –7

c 4 – (–3)



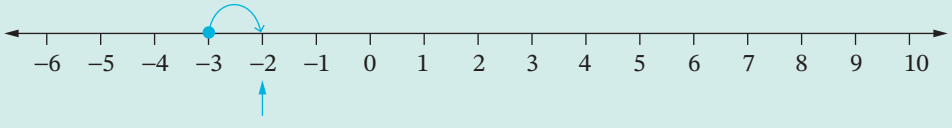
- 4 is where you start
- – direction is left
- –3 is how far you move in the *opposite* direction (that is, right)

4 – (–3) = 7

4 – (–3) is the same as 4 + 3

Subtracting a negative integer is like adding its opposite.

d –3 – (–1)



- –3 is where you start
- – direction is left
- –1 is how far you move in the opposite direction (that is, right)

–3 – (–1) = –2

–3 – (–1) is the same as –3 + 1

Worksheet
Adding and subtracting integers

Skillsheets
Integers

Integers using coloured squares

Interactives
Number line: Operations

Number line: Distance

1.05

Subtracting integers

When subtracting integers on a number line:

- subtracting a **positive** integer means moving to the left
- subtracting a **negative** integer is the same as **adding its opposite**, so it means moving to the right: $4 - (-3) = 4 + 3$

Note: '4 – (–3)' is read as '4 minus negative 1', not '4 minus minus 1'.

EXERCISE 1.05 ANSWERS ON P. 562

Subtracting integers U F R C

EXAMPLE 3

1 Evaluate each difference.

a	$4 - 2$	**b**	$2 - 4$	**c**	$3 - 13$	**d**	$5 - (-1)$
e	$7 - (-6)$	**f**	$-3 - (-2)$	**g**	$-5 - (-5)$	**h**	$-5 - 5$
i	$-6 - 3$	**j**	$-4 - (-5)$	**k**	$2 - 9$	**l**	$4 - (-2)$
m	$-6 - (-1)$	**n**	$12 - 18$	**o**	$9 - (-3)$	**p**	$-2 - 7$

2 What is the correct answer for $7 - 12$? Select the correct answer **A**, **B**, **C** or **D**.

A	19	**B**	-5	**C**	-19	**D**	5

3 **a** Is the answer to $10 - 4$ the same as the answer to $4 - 10$?

b Is $-5 - 9$ the same as $9 - (-5)$?

c Can integers be subtracted in any order? R C

4 Evaluate each expression.

a	$20 - 9$	**b**	$9 - 20$	**c**	$20 - (-9)$	**d**	$-9 - 20$
e	$-4 - 27$	**f**	$-4 - (-27)$	**g**	$13 - (-3)$	**h**	$-13 - (-3)$
i	$0 - 50$	**j**	$0 - (-50)$	**k**	$17 - (-25)$	**l**	$-17 - 25$
m	$16 - 8 - 20$	**n**	$-4 - (-10) - 1$				

5 What happens when you subtract an integer from its opposite? R C

6 For each difference, state whether it is always positive (+), always negative (–), or could be either positive or negative (E). R C

a A larger integer minus a smaller integer

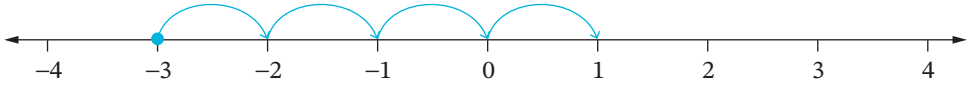
b A smaller integer minus a larger integer

c A positive integer minus a positive integer

d A positive integer minus a negative integer

e A negative integer minus a negative integer

7 The difference between 1 and –3 is also how much 1 is more than –3 on the number line. R



a Find the difference between 1 and –3.
b Evaluate 1 – (–3).
c Find the difference between 4 and 1.
d Evaluate 4 – 1.
e Find the difference between –1 and –2.
f Evaluate –1 – (–2).
g Find the difference between –1 and 3.
h Evaluate –1 – 3.
i Find the difference between –4 and –2.
j Evaluate –4 – (–2).

8 Copy and complete the blank for each equation. R

a 4 – ___ = –3
b –1 – ___ = –4
c –2 – ___ = –10
d ___ – (–1) = 9
e ___ – 6 = –5
f ___ – (–4) = –2

9 Copy and complete the blanks for each equation, where at least one of the numbers must be negative. R

a ___ – ___ = 3
b ___ – ___ = –2
c ___ – ___ = 0
d ___ – ___ = –5
e ___ – ___ = 6
f ___ – ___ = –11

Investigation

Multiplying integers

1 Copy and complete each pattern.

a 9, 6, 3, 0, –3, ___, ___, ___
b 4, 2, 0, ___, ___, ___
c 15, 10, 5, ___, ___, ___

2 Copy and complete each multiplication.

a 3 × (–7) = (–7) + (–7) + (–7) = ____
b 2 × (–5) = (–5) + (–5) = ____
c 3 × (–4) = (–4) + (–4) + (–4) = ____
d 4 × (–2) = (–2) + (–2) + (–2) + (–2) = ____
e 2 × (–6) = ____
f 3 × (–3) = ____

Foundation Standard Complex

3 **a** In pairs, or as a group activity, copy the multiplication grid below onto a large piece of paper and complete the shaded section.

×	5	4	3	2	1	0	−1	−2	−3	−4	−5
5											
4											
3											
2											
1											
0											
−1											
−2											
−3											
−4											
−5											

b Continue the pattern for the first row: 25, 20, 15, 10, 5, 0, −5, ...

c Continue the pattern for the second row: 20, 16, 12, 8, 4, 0, −4, ...

d Complete the next 4 rows.

e Continue the pattern for each **column**.

4 Use your completed table to evaluate each product.

a $4 \times (-3)$ **b** -3×5 **c** $-4 \times (-2)$ **d** $5 \times (-1)$

e $0 \times (-4)$ **f** -3×3 **g** $2 \times (-4)$ **h** $-5 \times (-5)$

5 How do the signs (positive or negative) of the integers in the question affect the sign of the product (answer)? Try to put the rules for multiplying integers into words and write them down.

1.06 Multiplying integers

Puzzle
Multiplying integers game

Worksheet
Integers writing activity

Skillsheet
Integers

Multiplying integers

When multiplying integers:

positive × positive = positive

positive × negative = negative

negative × positive = negative

negative × negative = positive

×	+	−
+	+	−
−	−	+

or

- If both numbers have the **same** sign, the answer is **positive**.
- If both numbers have **different** signs, the answer is **negative**.

Example 4

Evaluate each product:

a -3×2 **b** $-6 \times (-7)$ **c** $4 \times (-4)$ **d** $(-6)^2$

A **product** is the answer to a multiplication.

Solution

a $-3 \times 2 = -6$ negative × positive = negative

b $-6 \times (-7) = 42$ negative × negative = positive

c $4 \times (-4) = -16$ positive × negative = negative

d $(-6)^2 = (-6) \times (-6) = 36$ negative × negative = positive

$(-6)^2$ is '–6 squared' and means multiplying –6 by itself.

Video
Multiplying and dividing integers

Spreadsheet
Integer quiz: Multiplication

1.06

EXAMPLE 4

EXERCISE 1.06 ANSWERS ON P. 562

Multiplying integers UFRC

1 Evaluate each product.

a -3×6 **b** $6 \times (-4)$ **c** $-3 \times (-7)$

d $4 \times (-8)$ **e** $-7 \times (-9)$ **f** -9×5

g 7×6 **h** $-7 \times (-6)$ **i** $3 \times (-8)$

j -6×1 **k** $-1 \times (-1)$ **l** -9×6

m $5 \times (-2)$ **n** -7×6 **o** $-11 \times (-4)$

p -9×9 **q** $-12 \times (-5)$ **r** $20 \times (-3)$

s $-10 \times (-10)$ **t** $-7 \times (-8)$

2 For each expression, select the correct answer **A**, **B**, **C** or **D**.

a $8 \times (-5)$

A 13 **B** –13 **C** –40 **D** 40

b $-7 \times (-3)$

A 21 **B** –21 **C** –10 **D** 10

c $-2 \times 3 \times 7$

A 1 **B** 19 **C** –35 **D** –42

3 Copy and complete each multiplication grid.

a

×	–3	5
–3		
5		

b

×	–5	5
5		
–5		

c

×	6	–4
–4		
6		

d

×	–1	1
–1		
1		

e

×	7	–3
4		
–6		

f

×	–3	–6
8		
2		

Foundation Standard Complex

4 **a** Is the answer to $6 \times (-4)$ the same as the answer to $(-4) \times 6$?

b Can integers be multiplied in any order? R C

5 What happens when you multiply an integer by its opposite? R C

6 Evaluate:

a 3^2 **b** $(-3)^2$ **c** 7^2 **d** $(-7)^2$

e 5^2 **f** $(-5)^2$ **g** 1^2 **h** $(-1)^2$

7 For each product, state whether it is always positive (+), always negative (–), or could be either positive or negative (E). R C

a The product of 2 positive integers

b The product of 2 negative integers

c The product of a positive integer and a negative integer

d A positive integer squared

e A negative integer squared

8 Copy and complete the blank for each equation.

a $2 \times ___ = -10$ **b** $-4 \times ___ = 16$ **c** $9 \times ___ = -27$

d $___ \times (-5) = 35$ **e** $___ \times 8 = -40$ **f** $___ \times (-11) = 22$

9 Copy and complete the blanks for each equation, where at least one of the numbers must be negative.

a $___ \times ___ = 18$ **b** $___ \times ___ = -28$ **c** $___ \times ___ = 30$

d $___ \times ___ = -2$ **e** $___ \times ___ = 6$ **f** $___ \times ___ = -45$

Technology

Daily temperatures in Thredbo

This spreadsheet shows the minimum and maximum daily temperatures for Thredbo for the last 2 weeks of July.

1 Enter the data into a spreadsheet.

2 In cell D1, enter the label 'Difference'. To find the difference between the maximum and minimum temperatures for Sunday 18, enter the formula '=C2-B2' in cell D2.

	A	B	C	D
1	Date	Min Temp (deg C)	Max Temp (deg C)	Difference
2	Sun 18	-5	4.8	9.8
3	Mon 19	0	4	
4	Tue 20	-8.5	5	
5	Wed 21	-9.5	5.3	
6	Thu 22	-5	4.9	
7	Fri 23	-1.2	7	
8	Sat 24	-4.5	8	
9	Sun 25	-4	6.8	
10	Mon 26	-6.4	7	
11	Tue 27	-8	8.5	
12	Wed 28	-6	7.5	
13	Thu 29	-3.5	7.4	
14	Fri 30	1.6	8.5	
15	Sat 31	3	6	
16				

Foundation Standard Complex

3 To copy D2's formula into cells D3 to D15, click on D2 and position the cursor over the bottom right-hand corner of this cell so that it becomes a '+' sign. Drag down to cell D15 to highlight cells D3 to D15, as shown on the previous page. This is called **Fill Down** and you should notice that the temperature differences for each day have been calculated. Fill Down can also be selected by highlighting cells D2 to D15 and either selecting Fill Down from the Editing toolbar in the Home menu or by pressing Ctrl-D.

4 **a** On which day was the largest difference between the maximum and minimum temperatures recorded? Enter your answer in cell E1.

b On which day was the smallest difference recorded? Enter your answer in cell E2.

5 Enter the following labels: 'Lowest Minimum' in cell A17, 'Highest Minimum' in cell A18, 'Lowest Maximum' in D17, 'Highest Maximum' in D18, 'Difference' in A19.

6 **a** To find the lowest minimum, in cell B17 enter the formula '**=min(B2:B15)**', which finds the lowest number from cells B2 to B15.

b To find the highest minimum, in cell B18 enter '**=max(B2:B15)**'.

c Now, enter appropriate formulas into cells C17 and C18 for the lowest and highest maximums.

d What do you notice about the value in cell B17, compared to the other 3 values?

7 In cell B19 enter '**=B18-B17**' for the difference between the highest and lowest minimums. From cell B19, use **Fill Right** to copy this formula into cell C19. Alternatively, you can highlight cells B19 and C19 and either select Fill Right from the Editing toolbar in the Home menu or press Ctrl-R.

8 **a** Which column (B or C) had the greatest difference in temperature? Enter your answer in cell E3.

b In cell E4, enter a formula to find the difference between the lowest minimum temperature and the highest maximum temperature.



Alamy Stock Photo/Sindre Ellingsen

Investigation

Dividing integers

1 In pairs or as a group activity, copy and complete each question.

a	$5 \times ___ = -20$	**b**	$-6 \times ___ = 12$	**c**	$-3 \times ___ = 27$
d	$___ \times (-4) = -16$	**e**	$___ \times 2 = -10$	**f**	$___ \times (-5) = 15$
g	$-20 \div 5 = ___$	**h**	$27 \div (-3) = ___$	**i**	$-16 \div (-4) = ___$
j	$-14 \div 2 = ___$	**k**	$-12 \div (-12) = ___$	**l**	$21 \div (-3) = ___$

2 Write down the rules for dividing integers in words.

1.07 Dividing integers

Worksheets
Integers review

What is the question?

Skillsheet
Integers

Because division is the opposite of multiplication, the rules for dividing integers are the same as those for multiplying them.

Dividing integers

When dividing integers:

- positive ÷ positive = positive
- positive ÷ negative = negative
- negative ÷ positive = negative
- negative ÷ negative = positive

÷	+	−
+	+	−
−	−	+

or

- If both numbers have the **same** sign, the answer is **positive**.
- If both numbers have **different** signs, the answer is **negative**.

Video
Multiplying and dividing integers

Example 5

Evaluate each quotient:

a $-15 \div 3$ **b** $18 \div (-2)$ **c** $-24 \div (-4)$ **d** $\frac{14}{-7}$

A **quotient** is the answer to a division.

Solution

a	$-15 \div 3 = -5$	negative ÷ positive = negative
b	$18 \div (-2) = -9$	positive ÷ negative = negative
c	$24 \div (-4) = 6$	negative ÷ negative = positive
d	$\frac{14}{-7} = 14 \div (-7) = -2$	positive ÷ negative = negative

EXERCISE 1.07 ANSWERS ON P. 562

Dividing integers U F R C

EXAMPLE 5

1.07

1 Evaluate each quotient.

a	$20 \div (-5)$	**b**	$-8 \div 4$	**c**	$-12 \div (-3)$	**d**	$6 \div (-2)$
e	$27 \div (-3)$	**f**	$-18 \div (-6)$	**g**	$24 \div (-8)$	**h**	$40 \div (-4)$
i	$-20 \div 5$	**j**	$-21 \div (-3)$	**k**	$45 \div (-5)$	**l**	$-36 \div 6$
m	$-45 \div (-5)$	**n**	$-16 \div 2$	**o**	$28 \div (-7)$	**p**	$48 \div (-8)$

2 For each expression, select the correct answer **A**, **B**, **C** or **D**.

a $-48 \div (-2)$

A 24 **B** -24 **C** 46 **D** -46

b $\frac{16}{-4}$

A 12 **B** -4 **C** 4 **D** -12

3 Evaluate:

a	$\frac{12}{-2}$	**b**	$\frac{100}{-50}$	**c**	$\frac{-18}{-3}$	**d**	$\frac{-32}{-2}$
e	$\frac{-42}{6}$	**f**	$\frac{45}{-5}$	**g**	$\frac{100}{5}$	**h**	$\frac{-81}{-9}$

4 Divide the top row by the left-hand column to complete each division grid.

a

÷	−6	−3	3	6
−6				
−3				

b

÷	8	−8	32	−32
−2				
4				

5 What happens when you divide an integer by its opposite? R C

6 For each quotient, state whether it is always positive (+), always negative (−), or could be either positive or negative (E). R C

a The quotient of 2 positive integers

b The quotient of 2 negative integers

c The quotient of a positive integer and a negative integer

Foundation Standard Complex

7 Copy and complete the blank for each equation.

a $16 \div ___ = -2$ **b** $-24 \div ___ = -8$ **c** $35 \div ___ = -5$

d $___ \div (-11) = 4$ **e** $___ \div 5 = -10$ **f** $___ \div (-8) = 4$

8 Copy and fill in the blanks for each equation, where at least one of the numbers must be negative.

a $___ \div ___ = 10$ **b** $___ \div ___ = -4$ **c** $___ \div ___ = 5$

d $___ \div ___ = -8$ **e** $___ \div ___ = 6$ **f** $___ \div ___ = -3$

1.08 Order of operations

Videos
Order of operations
BODMAS

Now that you can add, subtract, multiply and divide with integers, you can evaluate **mixed expressions** involving integers. Remember to follow the **'order of operations'** rules.

Order of operations

1 Evaluate any expression inside **grouping symbols** (brackets), or in the numerator (top) and denominator (bottom) of a fraction

2 Evaluate any multiplication (×) and division (÷), from left to right

3 Evaluate any addition (+) and subtraction (–), from left to right

Skillsheet
Order of operations

Puzzles
Order of operations puzzle
Directed numbers

Example 6

Evaluate each expression.

a $3 \times (-4 - 6)$ **b** $8 + (-7) \times 4 - (-2)$

c $\dfrac{-13 + 5}{10 - 2}$ Imagine 'invisible brackets' around the top and the bottom of the fraction.

Solution

a $3 \times (-4 - 6) = 3 \times (-10)$ brackets first: $-4 - 6 = -10$

$= -30$

b $8 + (-7) \times 4 - (-2) = 8 + (-28) - (-2)$ multiplication first: $(-7) \times 4 = -28$

$= -20 - (-2)$ left to right: $8 + (-28) = -20$

$= -18$ $-20 - (-2) = -20 + 2$

c $\dfrac{-13 + 5}{10 - 2} = \dfrac{-8}{8}$ for fractions, numerator and denominator separately

$= -1$ $-8 \div 8 = -1$

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EXERCISE 1.08 ANSWERS ON P. 562

Order of operations U F R C

1 Evaluate each expression.

a $9 - 12 + 10$ **b** $-13 - 10 - (-1)$ **c** $4 - 6 + 2 + (-1)$

d $1 - 6 + 7 - 9$ **e** $6 + (-3) - 9$ **f** $7 - 5 - (-4) + (-2)$

g $5 \times (-10) \div 2$ **h** $6 \times (-4) \div (-3)$ **i** $7 \times 6 \div (-2)$

j $-3 \div (-1) \times 4$ **k** $24 \div (-6) \times 50$ **l** $-56 \div (-8) \times 4$

m $-100 \div (-20) \times 3$ **n** $-5 \times (-9) \div 9$ **o** $2 \times 12 \div (-4) \div (-2)$

EXAMPLE 6

2 What is the value of $6 + 4 \times (-2)$? Select the correct answer **A**, **B**, **C** or **D**.

A 14 **B** −20 **C** −2 **D** −14

3 Evaluate each expression.

a $-5 \times 2 + 3$ **b** $-5 \times (2 + 3)$ **c** $3 \times 6 \times (-2)$

d $12 - 8 \div 4$ **e** $12 - 8 \div (-4)$ **f** $-3 \times (9 - 10)$

g $21 \div (-7) + 8$ **h** $[-6 - (-2)] \times (-4)$ **i** $(-6 - 2) \div (-4)$

j $-3 \div 3 + 9$ **k** $12 \times (6 - 7) + 1$ **l** $-4 \times 7 \div (-7 - 7)$

m $\frac{25 - 7}{6}$ **n** $\frac{7 + (-3)}{-2 + 6}$ **o** $\frac{5 - 11}{12 - 10} + 6$

4 Copy and complete the blank for each equation. R

a $-5 \times 6 - ___ = -38$ **b** $___ \times (6 - 8) = -10$

c $[8 - ___] \times 4 = 44$ **d** $5 + ___ \times (-7) = 19$

e $8 - (-3 \times ___) = 17$ **f** $___ \times (-5 + 3) = -24$

g $6 \times ___ + (-4) = -16$ **h** $15 \div (-3) \times ___ = 10$

i $___ + 4 \times 6 = 5$ **j** $[-6 - ___] \times (-2) = -6$

k $(-6 - 3) \div ___ = -3$ **l** $___ \div (-5) + 6 = 1$

5 Nuraan completed these 10 questions. Mark her work and correct any mistakes.

a $6 - 9 + 5 - 11 = -9$ **b** $-32 \div (-4) \times (-2) = 4$

c $-6 \times (-2) - 8 = -4$ **d** $-11 + 11 \div (-1) = -22$

e $24 \div (-4) \div (-2) = 3$ **f** $(-2 + 6) \times 2 = -8$

g $-7 - 6 + 10 = 3$ **h** $(-12 - 3) \div (-3) = 3$

i $12 - 3 \times 5 = 45$ **j** $2 - 9 \times (-3) - (-18) = 39$

6 Copy each equation and insert grouping symbols to make it true. R C

a $-4 \times 7 - 2 + (-3) = -8$ **b** $-4 \times 7 - 2 + (-3) = -27$

c $-4 \times 7 - 2 + (-3) = -33$ **d** $40 \div 10 \times (-2) + 15 = 13$

e $40 \div 10 \times (-2) + 15 = 52$ **f** $40 \div 10 \times (-2) + 15 = -8$

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7 Evaluate $-24 \div 3 + (-4 + 7) \times (-2)$. Select **A**, **B**, **C** or **D**.

A 10 **B** 8 **C** −14 **D** −26

8 Copy each equation and use all the integers 4, −8 and 3 and grouping symbols to complete the equation. R C

a _____ = −20 **b** _____ = 15

c _____ = 8

Quiz
Mental skills 1B

Mental skills 1B: Maths without calculators ANSWERS ON P. 563

Subtracting 8 or 9

A quick way to mentally subtract 9 or 8 from a number is to subtract 10 and count forward 1 or 2 respectively.

1 Study each example.

a $66 - 9 = 66 - 10 + 1$
$= 56 + 1$
$= 57$
Count: '66, 56, 57'

b $83 - 8 = 83 - 10 + 2$
$= 73 + 2$
$= 75$
Count: '83, 73, 75'

c $72 - 49 = 72 - 50 + 1$
$= 22 + 1$
$= 23$
Count: '72, 22, 23'

d $141 - 28 = 141 - 30 + 2$
$= 111 + 2$
$= 113$
Count: '141, 111, 113'

2 Now evaluate each difference by counting.

a	26 − 9	**b**	44 − 9	**c**	123 − 9	**d**	270 − 9
e	161 − 29	**f**	187 − 59	**g**	75 − 19	**h**	457 − 39
i	82 − 8	**j**	131 − 8	**k**	96 − 8	**l**	120 − 8
m	44 − 8	**n**	577 − 28	**o**	203 − 18	**p**	365 − 48

1.09 Integers and the calculator

Negative numbers can be entered into a calculator using the sign change key (−) or +/−.
For example, to enter '−3', type (−) 3 or +/− 3.

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Example 7

Use your calculator to evaluate:

a $798 \div (-7) + 135$ **b** $23 \times [10 - (-6)]$ **c** $(-18)^2$

Solution

a $798 \div (-7) + 135 = 21$ On a calculator, enter 798 ÷ (–) 7 + 135 =

b $23 \times [10 - (-6)] = 368$ On a calculator, enter 23 × (10 – (–) 6) =

c $(-18)^2 = 324$ On a calculator, enter ((–) 18) x^2 =

There are more examples in Calculator Skills at the back of this book on page 555.

EXERCISE 1.09 ANSWERS ON P. 563

Integers and the calculator U F R C

1 Evaluate each expression *without* using a calculator.

a $6 - 13$ **b** $4 - (-2)$ **c** $-7 + 3$ **d** $-5 - (-6)$

e $3 + (-8)$ **f** $11 - (-3)$ **g** $11 - (-15)$ **h** $-13 + (-15)$

i $12 - (-18)$ **j** -8×6 **k** $-11 \times (-4)$ **l** $-5 \times (-6)$

m $13 \times (-5)$ **n** $88 \div (-11)$ **o** $-30 \div (-2)$ **p** $-36 \div 9$

q $(-5)^2$ **r** $(-8)^2$ **s** $-3 \times (-4)^2$ **t** $18 \div (-3)^2$

2 Check your answers to question **1** using a calculator.

3 When using your calculator, why are there different answers for $(-5)^2$ and -5^2? Explain the difference in meaning between $(-5)^2$ and -5^2. R C

4 Use a calculator to evaluate each expression.

a $13 + (-22) - 9$ **b** $64 + 23 + (-16)$ **c** $-200 \div (-100) \div 2$

d $-40 \times (-3) \div (-20)$ **e** $-44 \times (-15) \times (-6)$ **f** $48 \div 3 \times (-12)$

g $125 \div (-5) \div (-5)$ **h** $[12 - (-10)] \times (-3) + 6$ **i** $-3 \times (-8) \div (-4) + (-11)$

j $9 - 19 - 8 \times (-6)$ **k** $-12 - 30 \div (-2) + 58$ **l** $6 + 4 \times (-5) \div (-2)$

5 Copy and complete the blank for each equation. R C

a $6 \times (___ - 27) = -84$ **b** $___ - 8 - (-14) = -4$ **c** $45 \div (-9) \times ___ = -10$

d $-2 \times ___ + 11 = -19$ **e** $___ \times (-4) \div 2 = 112$ **f** $-14 - 6 \times ___ = 34$

6 Copy each equation and use all the integers 75, –18 and –15 and grouping symbols to complete the equation.

a _____ $= -1395$ **b** _____ $= -13$ **c** _____ $= -855$

1.09

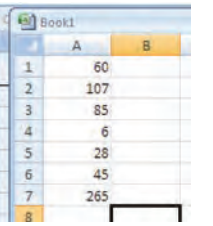
EXAMPLE 7

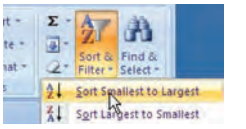
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Technology

Sorting numbers

In this activity, we will sort this set of numbers {60, 107, 85, 6, 28, 45, 265} by using a spreadsheet.





1 **a** Enter the 7 numbers shown below, in the given order, into column A of a spreadsheet.

b Highlight cells A1 to A7 and select **Sort & Filter** from the Editing menu on the top right-hand side. Choose the option shown for ascending order: **Smallest to Largest**.

c The data should now be sorted from the smallest number, 6 (in cell A1), to the largest number, 265 (in cell A7).

d A set of numbers can also be sorted in descending order: choose **Largest to Smallest**.

2 Now enter each of the following sets of numbers in the columns given, and sort in ascending order.

a {55, 89, 36, 21, 19, 4, 95} in column B

b {–17, 14, –90, 58, 0, 392, –6} in column C

c {4987, –4200, 8740, –5176, 2601, 713, –4810} in column D

d {16 101, 12 167, 10 010, 11 412, 10 107, 10 761, 11 214} in column E

3 Sort each column of data from question **2** in descending order.

1.10 Integer problems

Puzzle
Integers group clues

Example 8

The average temperature on the planet Mars is –63°C, while the average temperature on Jupiter is –150°C. What is the difference between these temperatures?

Solution

Write a number sentence to calculate the answer.

$-63 - (-150) = -63 + 150$

$= 87$

The difference in temperature is 87°C.

EXERCISE 1.10 ANSWERS ON P. 563

Integer problems U F PS R

For each question, write a number sentence to calculate the answer.

EXAMPLE 8

1.10

1 In Canberra, the temperature reached a maximum of 11°C. Overnight, it dropped to a minimum of –2°C. How many degrees did it drop? PS

2 Craig walked 16 kilometres east and then 20 kilometres west. How far is he from his starting point? PS

3 The heights above and below sea level of some places around the world are shown: PS

The Dead Sea (Jordan)	–397 metres
Death Valley (USA)	–86 metres
Mt Kosciuszko (Australia)	2230 metres
Mt Everest (Nepal)	8840 metres

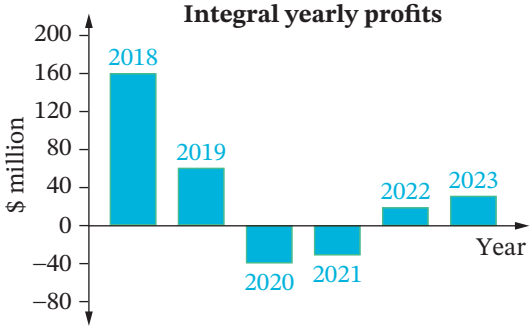
a How much higher is Mt Everest than Mt Kosciuszko?

b How much higher is Mt Kosciuszko than Death Valley?

c How much lower is the Dead Sea than Death Valley?

d The highest point on Earth is Mt Everest and the lowest point is the Dead Sea. What is the difference in height between these two points?

4 This graph shows the yearly profits made by the Integral company over 6 years. R



a In which years did Integral make a profit?

b In which years did Integral make a loss?

c What was the decrease in profit between 2019 and 2020?

d Did the company make an overall profit or loss over the 6 years shown? How much profit or loss did they make?

5 Anya left home and walked 3 kilometres south to Brock's home. Together, they then walked north for 4 kilometres. What distance and direction are they from Anya's home now? PS

6 Karina had $74 in her bank account yesterday but was allowed to withdraw $121. Today, she deposited $40 into the account. What is her account balance now? PS

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7 At Mt Hotham, the temperature overnight dropped to –6°C. During the day, it rose by 13°C, but then dropped 5°C after sunset. Which expression describes the temperature in Cooma after sunset? Select the correct answer **A**, **B**, **C** or **D**.

A $-6 + 13 - 5$ **B** $-6 - 13 - 5$

C $-6 + 13 + 5$ **D** $-5 - 13 - 6$

8 In indoor cricket, each time a batting player gets out, the team's score is reduced by 5 (that is, –5 is added to the total). Liam's cricket team is midway through a match. The first 5 batters have each scored 23, 16, 19, 0 and 16 before getting out. Liam is still batting and has scored 10 so far. What is the current score for Liam's team? PS

9 Mt Etna is an active volcano in Italy, 3200 m high. The lava starts 652 m below the Earth's surface and shoots to a height of 750 m above the top of Mt Etna. How far does the lava travel? PS



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10 A bird dives into a river from a height of 320 cm above the water, travelling at an average speed of 60 cm per second. PS

a How far does the bird travel in 8 seconds?

b How far below the water will it be after 8 seconds?

11 The submarine Nemesis descends from the surface to a depth of 955 metres to inspect a shipwreck. It then ascends 800 metres to send a message. How far below the surface is it when it sends the message? PS

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12 Find the closing balance of this bank account after all deposits and withdrawals. PS R

Starting balance	$37.85
Deposit	$18.20
Deposit	$120.00
Withdraw	$45.00
Deposit	$8.15
Withdraw	$248.20

13 A scuba diver was swimming at a depth of 18 metres. Her diving friend was at a depth of 7 metres. What was the vertical distance between them? PS

14 Jayden's bank balance is –$120, but he deposits $9 each day for the next 20 days. PS R

a On which day will his bank balance become positive?

b What will be his bank balance after 20 days?

Investigation

Above and below sea level

1 What is sea level?

2 Where is the highest point in Australia? What is its height above sea level?

3 Where is the lowest point in Australia? What is its height below sea level?

4 What is the difference in height between the highest and lowest points in Australia? Is it more than a kilometre?

5 What is the height of your town or city above sea level?

6 Find 5 places in the world that are above sea level and five that are below sea level. Write four questions about this information for other students to answer. Swap your work with other students and answer their questions.

Note: you must know the answers to your questions so that you can mark other students' work!

Power plus ANSWERS ON P. 563

1 Copy and complete each pattern.

a $(-1)^2 = (-1) \times (-1) =$ ___

b $(-1)^3 = (-1) \times (-1) \times (-1) =$ ___

c $(-1)^4 = (-1) \times (-1) \times (-1) \times (-1) =$ ___

d $(-1)^5 =$ _____ = ____

e $(-1)^6 =$ _____ = ____

f $(-1)^7 =$ _____ = ____

2 Explain how $(-1)^n$, where n is a whole number, is equal to only one of 2 possible numbers, and which one it is depends on the type of number n is.

3 Find the number (or numbers) that make each of these expressions true.

a $____^2 = 16$ **b** $____^2 = 100$ **c** $____^3 = -8$

4 Dylan says there is no answer to $____^2 = -25$. Is he correct?

5 **a** Find 3 integers that have a sum of –7.

b Find 2 negative integers that have a sum of –1.

c Find 3 integers that have a product of 36.

d Find 3 integers that have a product of –24.

6 Find the average of:

a –8 and 12 **b** –1 and –17 **c** 21 and –19

7 If a and b are integers, determine whether the following equations are true or false.

a $a + b = b + a$ **b** $a - b = b - a$ **c** $a \times b = b \times a$

d $a \div b = b \div a$ **e** $a + (-a) = 0$ **f** $(-a) \times (-a) = a \times a$

g $a - (-a) = 0$ **h** $a \div (-a) = -1$ **i** $(-a) + (-a) = -2 \times a$

8 Find 2 integers that have:

a a sum of –3 and a product of –10

b a sum of –8 and a product of 15

c a sum of –2 and a product of –8

d a sum of –1 and a product of –6

e a sum of 7 and a product of –30

CHAPTER 1 REVIEW

Language of maths

ascending	Celsius	degrees (°)	deposit
descending	difference	evaluate	greater than (>)
integer	less than (<)	minus	negative
number line	opposite	order	order of operations
positive	product	quotient	sign
sum	whole number	withdraw	zero

1 Find 4 pairs of words that are opposites.

2 What number is neither positive nor negative?

3 Write sentences using the words 'ascending' and 'descending' to show their meanings.

4 Write '$-5 < -1$' in words.

5 What is the order of operations?

6 What is the difference between -4 ('negative 4') and $- 4$ ('minus 4')?

Quiz
Language of maths 1

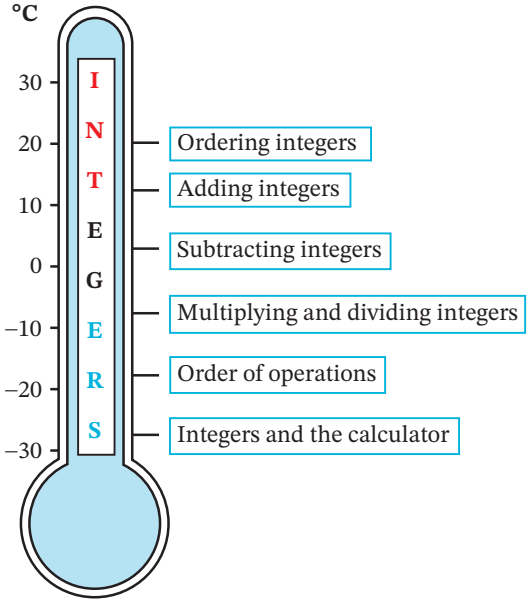
Puzzle
Integers find-a-word

Topic summary

- Using your own words, write down what integers are.
- Write any rules you have learnt about integers.
- What parts of this topic did you find difficult? What parts didn't you understand? Discuss them with a friend or your teacher.
- Give some examples of where negative numbers are used.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

Worksheet
Mind map: Integers



Quiz
Test yourself 1

TEST YOURSELF 1

ANSWERS ON P. 563

1.01

1 Write the opposite of each integer.

a −2 **b** −23 **c** 56
d −10 **e** 8 **f** 0

1.02

2 Use a separate number line to mark each set of numbers.

a 6, −3, 5, 1, −1, 4, 7
b integers between −2 and 5
c integers less than 1 and greater than −6

1.03

3 Write true (T) or false (F) for each statement.

a $-4 > 2$ **b** $-2 < -3$ **c** $-1 > 1$ **d** $2 > -1$
e $-11 < -10$ **f** $-3 > -4$ **g** $-8 < 11$ **h** $-4 > -3$
i $5 < -4$ **j** $-16 < 0$ **k** $1 > -5$ **l** $-9 > -2$

1.03

4 Write this set of integers in descending order:

−9, 14, 3, 0, −5, −10, −1.

1.04

5 Evaluate each expression.

a $3 + (-4)$ **b** $-8 + (-2)$ **c** $10 + (-13)$
d $-4 + 8$ **e** $-5 + (-5)$ **f** $-3 + 6$
g $-5 + (-4) + (-2)$ **h** $-8 + 5 + 4$ **i** $-7 + 6 + (-7)$

1.05

6 Evaluate each expression.

a $3 - (-4)$ **b** $-5 - (-9)$ **c** $7 - 11$
d $-4 - 8$ **e** $-5 - (-5)$ **f** $-3 - 4 - 5$
g $-8 + 5 - 4$ **h** $8 - 6 - (-4)$ **i** $3 - 8 + (-6)$

1.06

7 Evaluate each product.

a $6 \times (-3)$ **b** $-8 \times (-6)$
c -2×5 **d** -11×4
e $7 \times (-9)$ **f** 6×8
g $-6 \times (-4) \times (-1)$ **h** $-2 \times 3 \times 7$
i $-4 \times 2 \times (-5)$

1.07

8 Evaluate each quotient.

a $-12 \div (-4)$ **b** $15 \div (-5)$
c $-6 \div (-6)$ **d** $16 \div (-4)$
e $-24 \div 12$ **f** $-1000 \div 10$
g $24 \div (-6) \div (-2)$ **h** $-36 \div 3 \div 4$
i $-64 \div (-4) \div 2$

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9 Evaluate each expression. 1.08

a $-6 + (-3) \times 2$

b $4 - 3 \times 6$

c $-12 \div 4 - 10$

d $-3 \times 5 + 6 \div (-2)$

e $30 \div (-6) + 5$

f $-2 \times 7 + 15 \div (-3)$

g $-12 + (-3) \times (-7) - 2$

h $-6 + (-6) \div (-2) \times 2$

i $\frac{9 - 17}{-4 + 6}$

j $7 + \frac{-12 + 6}{2 - 5}$

10 Use your calculator to evaluate each expression. 1.09

a $28 - 37$

b $-17 + 11$

c $18 - 15 \times 2$

d $100 - 120 \div 4 + 17$

e $-16 \times 3 + (-4)$

f $[5 + (-20)] \div (-3)$

11 The minimum overnight temperature in Goulburn was –5°C. During the day, it rose to 11°C. By how much did the temperature rise? 1.10



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12 Harrison starts a new job and opens a new bank account. Each week he saves $80 to put into the account. Each month, however, $210 is withdrawn from the account to pay off a car loan. Calculate how much will be in Harrison's account after 6 months if there are 26 weeks in 6 months. 1.10

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